18.2: Eukaryotic gene expression is regulated at many stages

All organisms, whether prokaryotes or eukaryotes, must regulate which genes are expressed at any given time. Both unicellular organisms and the cells of multicellular organisms continually turn genes on and off in response to signals from their external and internal environments. Regulation of gene expression is also essential for cell specialization in multicellular organisms, which are made up of different types of cells. To perform its own distinct role, each cell type must maintain a specific program of gene expression in which certain genes are expressed and others are not (see Figure 18.1).

Differential Gene Expression

A typical human cell might express about a third to a half of its protein-coding genes at any given time. Highly differentiated cells, such as muscle or nerve cells, express a smaller fraction of their genes. Almost all the cells in a multicellular organism contain an identical genome. (Cells of the immune system are one exception, as you will see in Figure 43.13.) A subset of genes is expressed in each cell type; some of these—about 35%—are “housekeeping” genes, expressed by many cell types, while others are unique to that cell type. The uniquely expressed genes allow these cells to carry out their specific function. The differences between cell types, therefore, are due not to different genes being present, but to **differential gene expression**, the expression of different genes by cells with the same genome.

The function of any cell depends on its expressing the appropriate set of genes. The transcription factors of a cell must locate the right genes at the right time, a task like finding a needle in a haystack. Abnormal gene expression can cause serious imbalances and diseases, including cancer.

Figure 18.6 summarizes the process of gene expression in a eukaryotic cell, highlighting key stages in the expression of a protein-coding gene. Each stage depicted in Figure 18.6 is a potential control point at which gene expression can be turned on or off, accelerated, or slowed down. Many genes have more than one control point.

Figure 18.6

```
    Stages in gene expression that can be regulated in
eukaryotic cells.
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Stages in gene expression that can be regulated in eukaryotic cells.

Regulation of gene expression within the nucleus. After the cell receives a signal, gene expression can be regulated in the following processes:

- 1. Chromatin modification: D N A unpacking (which flows to gene in the D N A available for transcription).
- 2. Transcription results in the primary transcript R N A, with both introns and exons.
- 3. R N A processing: introns removed, cap and tail in place on R N A.
- 4. Transport to the cytoplasm, when m R N A exits the nucleus.

Regulation of gene expression in the cytoplasm. Gene expression can be regulated during the following processes:

- 1. Translation, which leads to the polypeptide chain.
- 2. Degradation of the m R N A (m R N A show here in pieces).
- 3. Protein processing, which results in an active protein (with functional shape).
- 4. Degradation of protein (protein shown here in pieces).
- 5. Transport to cellular destinations for cellular function (such as enzymatic activity or structural support).

In this diagram, the colored boxes indicate the processes most often regulated; each color indicates the type of molecule that is affected blue=DNA,red/orange=RNA,purple=protein. The nuclear envelope separating transcription from translation in eukaryotic cells allows for RNA processing, a form of post-transcriptional control that is not possible in prokaryotes. Eukaryotes also have a greater variety of control mechanisms operating before transcription and after translation. Some genes are regulated at multiple stages.

Animation

Control of Gene Expression

image

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image

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When the structure of DNA was determined in 1953, an understanding of the mechanisms that control gene expression in eukaryotes seemed almost hopelessly out of reach. Since then, advances in DNA technology (see Chapter 20) have enabled molecular biologists to uncover many details of eukaryotic gene regulation. In all organisms, gene expression is commonly controlled at transcription; regulation at this stage often occurs in response to signals coming from outside the cell, such as hormones or other signaling molecules. For this reason, the term *gene expression* is often equated with transcription for both bacteria and eukaryotes. While this may be the case for bacteria, the greater complexity of eukaryotic cell structure and function provides opportunities for regulating gene expression at many stages besides transcription (see Figure 18.6). Let's examine some of the important control points of eukaryotic gene expression more closely.

Regulation of Chromatin Structure

Recall that the DNA of eukaryotic cells is packaged with proteins in an elaborate complex known as chromatin, the basic unit of which is the nucleosome (see Figure 16.23). The structural organization of chromatin not only packs a cell's DNA into a compact form that fits inside the nucleus, but also helps regulate gene expression in several ways. Genes within heterochromatin, which is more densely arranged than euchromatin, are usually not expressed. In euchromatin, whether or not a gene is transcribed is affected by the location of nucleosomes along a gene's promoter and also the sites where the DNA attaches to the protein scaffolding of the chromosome. Chromatin structure and gene expression can be influenced by chemical modifications of both the histone proteins of the nucleosomes around which DNA is wrapped and the nucleotides that make up that DNA. Here we examine the effects of these modifications, which are catalyzed by specific enzymes.

Histone Modifications and DNA Methylation

Chemical modifications to histones, found in all eukaryotic organisms, play a direct role in the regulation of gene transcription. The N-terminus of each histone protein in a nucleosome protrudes outward from the nucleosome **(Figure 18.7a)**. These so-called *histone tails* are accessible to various modifying enzymes that catalyze the addition or removal of specific chemical groups, such as acetyl (—$COCH_3$), methyl, and phosphate groups (see Figure 4.9). Generally, **histone acetylation**—the addition of an acetyl group to an amino acid in a histone tail—appears to promote transcription by opening up chromatin structure **(Figure 18.7b)**, while the addition of methyl groups to histones can lead to the condensation of chromatin and

reduced transcription. Often, the addition of a particular chemical group may create a new binding site for enzymes that further modify chromatin structure.

Figure 18.7

```
    A simple model of histone tails and the effect of histone
acetylation.
```

image A simple model of histone tails and the effect of histone acetylation.

Amino acids in histone tails may be chemically modified by addition of acetyl groups (green balls) or other groups (such as methyl or phosphate groups). Such modifications affect chromatin structure in a region, sometimes by providing binding sites for other chromatin-modifying enzymes.

Parts are as follows.

Part a. Histone tails protrude outward from a nucleosome. The amino acids in the histone tails are accessible for modification. Regulation of gene expression in this figure represents chromatin modification. Overview: Stages in gene expression subject to regulation detailed in a flow chart: Within nucleus: Chromatin modification, transcription, R N A processing; in the cytoplasm: m R N A degradation, translation and protein processing and degradation. Chromatin modification is highlighted for this figure. End view of a nucleosome is diagrammed, with D N A double helix wound around complex of four histone proteins. Histone tails protrude, making the amino acids in the tails available for chemical modification.

Part b. Acetylation of histone tails promotes loose chromatin structure that permits transcription. Regulation of gene expression in this figure represents chromatin modification. A region of chromatin in which nucleosomes are un-acetylated forms a compact structure in which the D N A is not accessible for transcription. In a side view of un-acetylated histones, the D N A can be seen as tightly wound around a compact structure of histones. When nucleosomes are highly acetylated (acetyl groups attached to protruding histone tails), the chromatin becomes less condensed, and the D N A is looser and accessible for transcription.

HHMI BioInteractive Animation

Role of MECP2 in Gene Silencing

image

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Rather than modifying histone proteins, a different set of enzymes can methylate the DNA itself on certain bases, usually cytosine. Such **DNA methylation** occurs in most plants, animals, and fungi. Long stretches of inactive DNA, such as that of inactivated mammalian X chromosomes (see Figure 15.8), are generally more methylated than regions of actively transcribed DNA. On a smaller scale, the DNA of individual genes is usually more heavily methylated in cells in which those genes are not expressed. Removal of the extra methyl groups can turn on some of these genes.

Once methylated, genes usually stay that way through successive cell divisions in a given individual. At DNA sites where one strand is already methylated, enzymes methylate the correct daughter strand after each round of DNA replication. Methylation patterns are thus passed on to daughter cells, and cells forming specialized tissues keep a chemical record of what occurred during embryonic development. A methylation pattern maintained in this way also accounts for *genomic imprinting* in mammals, where methylation permanently regulates expression of either the maternal or paternal allele of particular genes at the start of development (see Figure 15.17). DNA methylation and histone modification are believed to be coordinated in their regulation.

Epigenetic Inheritance

The chromatin modifications that we just discussed do not change the DNA sequence, yet they still may be passed along to future generations of cells. Inheritance of traits transmitted by mechanisms not involving the nucleotide sequence itself is called *epigenetic inheritance*, the study of which is called **epigenetics**. Whereas mutations in the DNA are permanent changes, modifications to the chromatin can be reversed. For example, DNA methylation patterns are largely erased during gamete formation and reestablished during embryonic development. Furthermore, they are changeable, thus responding more rapidly to environmental conditions.

Research on epigenetics has skyrocketed over the past 20 years The importance of epigenetic information in regulating gene expression is now widely accepted. One key study by researchers at Duke University in the early 2000s used a mouse mutant whose genome had been altered so that a gene called *agouti* that determines coat color, normally expressed only briefly during fur formation, was instead expressed throughout development. This overexpression resulted in yellow mice (**Figure 18.8a**, left mouse) rather than the usual brownish ("agouti") color. Previous work had shown that merely supplementing the diet of pregnant mothers with methyl group–containing compounds (like folic acid) could shift the range of coat colors of the offspring back to normal (see Figure 18.8a, right mouse). Waterland and Jirtle reproduced this result, analyzing the state of methylation of the DNA. They showed that the extent of the color shift

correlated with the level of DNA methylation. In other words, feeding methyl groups to the mothers at a key time during gestation led to a change in gene expression in the offspring's phenotype. Further studies showed that the effects were even observed in the next generation—the "grandpups" of the original female mouse.

Figure 18.8

Examples of epigenetic inheritance.

Part a. Effects of maternal diet on genetically identical mice.

Two genetically identical agouti mutant mice had mothers fed on different diets. A maternal diet without compounds that donate methyl groups resulted in an obese, yellow offspring. A diet supplemented with methyl-donating compounds, such as folic acid, led to a speckled brown fur mouse (normal agouti phenotype) and normal weight.

The Dutch hunger winter.

Part b. During the Nazi blockade of food transport into the Netherlands near the end of World War 2, the population was starving, including a featured mother and her two children. Children of mothers who were pregnant during that time have experienced long-term effects.

A similar epigenetic effect due to changes in methylation occurs in humans as well. Near the end of World War II, during the winter of 1944–45, Dutch railway workers went on strike to try to prevent the Nazis from bringing in more troops. In retaliation, the Nazis blocked all deliveries of food. Over 20,000 Dutch died in the "Dutch Hunger Winter" **(Figure 18.8b)**. Over time, doctors found that the offspring of individuals who were in early pregnancy at that time experienced adverse health effects as adults: higher rates of obesity, high triglyceride and cholesterol levels, type 2 diabetes, and schizophrenia. Furthermore, these affected offspring had a 10% higher mortality rate after the age of 68 than their siblings who were *in utero* when food was readily available. A collaboration between labs in the Netherlands and the United States compared those adults with their siblings and published their findings in 2018. Statistical analysis enabled the researchers to conclude that differences between the siblings in DNA methylation of certain genes caused these long-term adverse medical conditions—examples of epigenetic inheritance.

Epigenetic variations might help explain cases where one identical twin acquires a genetically based disease, such as schizophrenia, but the other does not, despite their identical genomes. Alterations in normal patterns of DNA methylation are seen in some cancers, where they are associated with inappropriate gene expression. Evidently, enzymes that modify chromatin structure are integral parts of the eukaryotic cell's machinery for regulating transcription.

Regulation of Transcription Initiation

Chromatin-modifying enzymes provide initial control of gene expression by making a region of DNA either more or less able to bind the transcription machinery. Once the chromatin of a gene is optimally modified for expression, the initiation of transcription is the next major step at which gene expression is regulated. As in bacteria, the regulation of transcription initiation in eukaryotes involves proteins that bind to DNA and either facilitate or inhibit binding of RNA polymerase. The process is more complicated in eukaryotes, however. Before looking at how eukaryotic cells control their transcription, let's review the structure of a eukaryotic gene.

Organization of a Typical Eukaryotic Gene and Its Transcript

A eukaryotic gene and the DNA elements (segments) that control it are typically organized as shown in **Figure 18.9**, which extends what you learned about eukaryotic genes in Chapter 17. Recall that a cluster of proteins called a *transcription initiation complex* assembles on the promoter sequence at the "upstream" end of the gene (see Figure 17.9). One of these proteins, RNA polymerase II, then proceeds to transcribe the gene, synthesizing a primary RNA transcript (pre-mRNA). RNA processing includes enzymatic addition of a 5′ cap and a poly-A tail, as well as splicing out of introns, to yield a mature mRNA. Associated with most eukaryotic genes are multiple **control elements**, segments of noncoding DNA that serve as binding sites for the proteins called transcription factors, which bind to the control elements and regulate transcription. Control elements on the DNA and the transcription factors that bind to them are critical to the precise regulation of gene expression seen in different cell types.

Figure 18.9

A eukaryotic gene and its transcript.

Regulation of gene expression: Transcription. Proximal and distal enhancers (gold) diagrammed.

Overview: Stages in gene expression subject to regulation detailed in a flow chart: Within nucleus: Chromatin modification, transcription, R N A processing; in the cytoplasm: m R N A degradation, translation and protein processing and degradation. Transcription is highlighted for this figure. Diagram of D N A molecule with regions labeled from left to right:

Enhancer, stretch with 3 distinct regions highlighted in gold (group of distal control elements).

Enhancer is upstream (left) and separated from another region with proximal control elements (two are highlighted here).

Just downstream (to the right) of the proximal control elements is a promoter with the transcription start sight at the downstream (far right) end.

Next (downstream) is a sequence of regions as follows: Exon, intron, exon, intron, exon, poly-A-signal sequence, and then, further downstream, the transcription termination region.

After transcription the primary R N A transcript is diagramed left to right: 5 prime, exon, intron, exon, intron, exon, poly-A signal, cleaved 3-prime end of primary transcript.

Each eukaryotic gene has a promoter—a DNA sequence where RNA polymerase II binds and starts transcription, proceeding "downstream." A number of control elements (gold) are involved in regulating the initiation of transcription; these are DNA sequences located near (proximal to) or far from (distal to) the promoter. Distal control elements can be grouped together as enhancers, one of which is shown for this gene. At the other end of the gene, a polyadenylation (poly-A) signal sequence in the last exon of the gene is transcribed into an RNA sequence that signals where the transcript is cleaved and the poly-A tail added. Transcription may continue for hundreds of nucleotides beyond the poly-A signal before terminating. RNA processing of the primary transcript into a functional mRNA involves three steps: addition of the 5′ cap, addition of the poly-A tail, and splicing. In the cell, the 5′ cap is added soon after transcription is initiated, and splicing occurs while transcription is still under way (see Figures 17.11 and 17.12).

The Roles of General and Specific Transcription Factors

There are two types of transcription factors: General transcription factors act at the promoter of all genes, while some genes require specific transcription factors that bind to control elements that may be close to or farther away from the promoter.

General Transcription Factors at the Promoter

To initiate transcription, eukaryotic RNA polymerase II requires the assistance of transcription factors. Some transcription factors (such as those illustrated in Figure 17.9) are essential for the transcription of *all* protein-coding genes; therefore, they are often called *general transcription factors*. A few general transcription factors bind to a DNA sequence, such as the TATA box in most promoters, but many bind to proteins, including other transcription factors as well as RNA polymerase II. Protein-protein interactions are crucial to the initiation of eukaryotic transcription. Only when the complete initiation complex has assembled can the polymerase begin to move along the DNA template strand and transcribe.

Some genes are expressed all the time, but others are not; instead, they are regulated. For these genes, the interaction of general transcription factors and RNA polymerase II with a promoter usually leads to a low rate of initiation and production of few RNA transcripts from genes that are not expressed at significant levels all the time or in all cells. In eukaryotes, high levels of transcription of these particular genes at the

appropriate time and place depend on the interaction of control elements with another set of proteins, which can be thought of as *specific transcription factors*.

Enhancers and Specific Transcription Factors

As you can see in Figure 18.9, some control elements, named *proximal control elements*, are close to the promoter. The more distant *distal control elements*, groupings of which are called **enhancers**, may be thousands of nucleotides upstream or downstream of a gene or even within an intron. A given gene may have multiple enhancers, each active at a different time, cell type, or location in the organism. Each enhancer, however, is generally associated with only that gene and no other.

In eukaryotes, the rate of gene expression can be strongly increased or decreased by the binding of specific transcription factors, either activators or repressors, to the control elements of enhancers. Hundreds of transcription activators have been discovered in eukaryotes; the structure of one is shown in **Figure 18.10**. Researchers have identified two types of structural domains that are commonly found in a large number of transcription activators: a *DNA-binding domain*—a part of the protein's three-dimensional structure that binds to DNA—and one or more *activation domains*. Activation domains bind other regulatory proteins or components of the transcription machinery, facilitating a series of protein-protein interactions that result in enhanced transcription of a given gene.

Figure 18.10

The structure of MyoD, a transcriptional activator.

The D N A binding domains of myo d attach to a double helix representing D N A. The activation domains of both subunits wind above the D N A molecule.

The MyoD protein is made up of two polypeptide subunits (purple and salmon) with extensive regions of α helix. Each subunit has one DNA-binding domain (lower half) and one activation domain (upper half). The latter includes binding sites for the other subunit and for other proteins. MyoD is involved in muscle development in vertebrate embryos (see Concept 18.4).

VISUAL SKILLS *Describe how the two functional domains of the MyoD protein relate to the two polypeptide subunits.*

*For suggested answer, see*Appendix A.**

Figure 18.11 shows the currently accepted model for how binding of activators to an enhancer located far from the promoter influences transcription. Protein-mediated bending of the DNA brings the bound activators into contact with a group of *mediator proteins*, which in turn interact with general transcription factors at the promoter. These protein-protein interactions help assemble and position the initiation

complex on the promoter. One study supporting this model shows that proteins regulating one of the mouse globin genes contact both the gene's promoter and an enhancer located about 50,000 nucleotides upstream. Protein interactions allow these two DNA regions to come together in a very specific fashion, in spite of the many nucleotide pairs between them.

Figure 18.11

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    A model for the action of enhancers and transcription
activators.
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A model for the action of enhancers and transcription activators.

Bending of the DNA by a protein enables enhancers to influence a promoter hundreds or even thousands of nucleotides away. Specific transcription factors (activators) bind to the enhancer DNA sequences and then to a group of mediator proteins. These in turn bind to general transcription factors and then RNA polymerase II, thus assembling the transcription initiation complex. These protein-protein interactions lead to correct positioning of the complex on the promoter and the initiation of RNA synthesis. Only one enhancer (with three gold control elements) is shown here, but a gene may have several enhancers that act at different times or in different cell types.

Parts are as follows.

1. Overview: Stages in gene expression subject to regulation detailed in a flow chart: Within nucleus: Chromatin modification, transcription, R N A processing; in the cytoplasm: m R N A degradation, translation and protein processing and degradation. Transcription is highlighted for this figure. 2. D N A with enhancer, promoter and gene. Activator proteins bind to distal control elements grouped in the D N A. This enhancer has three binding sites, each called a distal control element. Activator proteins shown bound to enhancer binding sites. Enhancer is far upstream (distant) from the promoter region, which includes the T A T A box (upstream,) and promoter. Just downstream of the promoter is a gene.
2. A D N A-bending protein brings the bound activators closer to the promoter by bending the D N A molecule into a U shape. General transcription factors, a group of mediator proteins, and R N A polymerase roman numeral 2 are nearby.
3. The activators bind to certain mediator proteins and general transcription factors, helping them form an active transcription initiation complex on the promoter. R N A polymerase 2 binds to the

complex at the promoter, and R N A synthesis begins downstream on the gene region (denoted by a red arrow).

Animation

Regulation of Transcription by Transcriptional Activators and Enhancers

image

Play

image

Play

Specific transcription factors that function as repressors can inhibit gene expression in several different ways. Some repressors bind directly to control element DNA (in enhancers or elsewhere), blocking activator binding. Other repressors interfere with the activator itself so it can't bind the DNA.

In addition to influencing transcription directly, some activators and repressors indirectly affect chromatin structure. Studies using yeast and mammalian cells show that some activators recruit proteins that acetylate histones near the promoters of specific genes, thus promoting transcription (see Figure 18.7). Similarly, some repressors recruit proteins that remove acetyl groups from histones, leading to reduced transcription, a phenomenon referred to as *silencing*. Indeed, recruitment of chromatin-modifying proteins seems to be the most common mechanism of repression in eukaryotic cells.

HHMI BioInteractive Animation

Regulation of Eukaryotic DNA Transcription

image

Play

image

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Combinatorial Control of Gene Activation

In eukaryotes, the precise control of transcription depends largely on the binding of activators to DNA control elements. Considering that many genes must be regulated in a typical animal or plant cell, the number of different nucleotide sequences in control elements is surprisingly small. A dozen or so short

nucleotide sequences appear again and again in the control elements for different genes. On average, each enhancer is composed of about ten control elements, each binding only one or two specific transcription factors. It is the particular *combination* of control elements in an enhancer associated with a gene, rather than a single unique control element, that is important in regulating transcription of the gene.

Even with only a dozen control element sequences available, many combinations are possible. Each combination of control elements can activate transcription only when the appropriate transcription activators are present, which may occur at a precise time during development or in a particular cell type. In the **Scientific Skills Exercise**, you can work with data from an experiment that identified the control elements in an enhancer of a particular human gene. **Figure 18.12** illustrates how the use of different combinations of just a few control elements can allow differential regulation of transcription in two representative cell types—liver cells and lens cells. This can occur because each cell type contains a different group of transcription activators. How cell types come to differ during this process, even though they all arise from one cell (the fertilized egg), will be explored in Concept 18.4.

Scientific Skills Exercise

Analyzing DNA Deletion Experiments

AP Science Practices 1, 2, 3, 4, 5, 6

What Control Elements Regulate Expression of the mPGES-1 **Gene?** The promoter of a gene includes the DNA immediately upstream of the transcription start site, but the control elements (grouped in an enhancer) that regulate the level of transcription of the gene may be thousands of base pairs upstream of the promoter. Because the distance and spacing of control elements make them difficult to identify, scientists begin by deleting possible control elements and measuring the effect on gene expression. In this exercise, you will analyze data obtained from DNA deletion experiments that tested possible control elements for the human gene mPGES-1. This gene codes for an enzyme that synthesizes a type of prostaglandin, a chemical made during tissue inflammation.

Image of a woman with inflammation in her shoulder.

Tissue inflammation

How the Experiment Was Done The researchers hypothesized that there were three possible control elements in an enhancer region located 8–9 kilobases (kb) upstream of the mPGES-1 gene. Control elements regulate whatever gene is in the appropriate downstream location. Thus, to test the activity of the possible

elements, researchers first synthesized molecules of DNA *(DNA constructs)* that had the intact enhancer region upstream of a *reporter gene*, a gene whose mRNA product could be easily measured experimentally. Next, they made three more DNA constructs, with one of the three proposed control elements deleted in each (see the left side of the figure). The researchers then introduced each DNA construct into a separate human cell culture, where the cells took up the DNA. After 48 hours, the amount of reporter gene mRNA made by the cells was measured. Comparing these amounts allowed researchers to determine if any of the deletions had an effect on expression of the reporter gene, mimicking the effect of deletions on mPGES-1 gene expression. (The mPGES-1 gene itself couldn't be used to measure expression levels because the cells express their own mPGES-1 gene, and the mRNA from that gene would confuse the results.)

Data from the Experiment The diagrams on the left side of the figure show the intact DNA sequence (top) and the three experimental DNA constructs. A red X is located on the possible control element (1, 2, or 3) that was deleted in each experimental DNA construct. The area between the slashes represents the approximately 8 kb of DNA located between the promoter and the enhancer region. The horizontal bar graph on the right shows the amount of reporter gene mRNA that was present in each cell culture after 48 hours relative to the amount that was in the culture containing the intact enhancer region (top bar=100%).

Diagram of four D N A sequences, each corresponding to a bar on a horizontal bar graph.

Data from J. N. Walters et al., Regulation of human microsomal prostaglandin E synthase-1 by IL-1b requires a distal enhancer element with a unique role for C/EBPb, *Biochemical Journal* 443:561–571 (2012).

Intact sequence has no enhancer deletions; 3 remaining sequences each have one possible control element in the enhancer deleted. Horizontal axis on the bar graph is relative level of reporter m R N A (percent of control). Range is from 0 to 200 with increments every 50. Intact D N A sequence is: Enhancer with possible control elements 1, 2 and 3; 8 kilobase region of separation; promoter; reporter gene. Corresponding bar on graph goes to 100. Second D N A sequence shows possible control element #1 deleted. Corresponding bar on graph goes to about halfway between 150 and 200. Third D N A sequence shows possible control element #2 deleted. Corresponding bar on graph goes to almost 150. Fourth D N A sequence has possible control element #3 deleted. Corresponding bar on graph goes to almost 50.

INTERPRET THE DATA

- ◦ What is the independent variable in the graph?
- What is the dependent variable?
- What was the control treatment in this experiment? Label it on the diagram.

- Do the data suggest that any of these possible control elements are actual control elements? Explain.
-
    - Did deletion of any of the possible control elements cause a *reduction* in reporter gene expression? If so, which one(s), and how can you tell?

- If loss of a control element causes a reduction in gene expression, what must be the normal role of that control element? Provide a biological explanation for how the loss of such a control element could lead to a reduction in gene expression.
-
    - Did deletion of any of the possible control elements cause an *increase* in reporter gene expression relative to the control? If so, which one(s), and how can you tell?

- If loss of a control element causes an increase in gene expression, what must be the normal role of that control element? Propose a biological explanation for how the loss of such a control element could lead to an increase in gene expression.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Figure 18.12

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Cell type–specific transcription.
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Cell type-specific transcription.

Both liver cells and lens cells have the genes for making the proteins albumin and crystallin, but only liver cells make albumin (a blood protein) and only lens cells make crystallin (the main protein of the lens of the eye). The specific transcription factors made in a cell determine which genes are expressed. In this example, the genes for albumin and crystallin are shown at the top, each with an enhancer made up of three different control elements. Although the enhancers for the two genes both have a gray control element, each enhancer has a unique combination of elements. All the transcription activator proteins required for high-level expression of the albumin gene are present in liver cells only (left), whereas the activators needed for expression of the crystallin gene are present in lens cells only (right). For simplicity, we consider only the role of specific transcription factors that are activators here, although repressors may also influence transcription in certain cell types.

Part a. Liver cell. D N A in both lens and liver cells contain the albumin gene and the crystalline gene. Diagram of D N A for these genes (regions labeled from left to right):

D N A molecule 1: Enhancer for albumin gene containing three control elements (here shown as gold, gray and orange); slashes representing a distance of some number of nucleotides followed downstream by the promoter region and the albumin gene.

D N A molecule 2: Enhancer for crystallin gene containing three control elements (here shown as brown, gray and pink); slashes representing a distance of some number of nucleotides followed downstream by the promoter region and the crystallin gene. In the liver cell nucleus, available activators bind to the control regions in the albumin gene enhancer, and a group of mediator proteins and a D N A-bending protein bend the D N A into a U so that the enhancer is near the promoter region for the albumin gene. R N A polymerase is able to bind to the promoter (completing the transcription initiation complex) and the albumin gene is expressed. D N A containing the crystallin gene is also present in the liver cell nucleus, but no activators are present to bind to the control regions in its enhancer and the crystallin gene is not expressed.

Part b. Lens cell nucleus. Activators for the control regions in the crystalline gene enhancer are present, but activators for the albumin gene control regions are not. Activators for the control regions in the crystallin gene enhancer bind, and the D N A bends with the help of D N A-bending protein and a group of mediator proteins, allowing R N A polymerase to bind as well (completing the transcription initiation complex) and the crystalline gene is expressed. D N A containing the albumin gene is also present in the lens cell nucleus, but no activators are present to bind to the control regions in its enhancer and the albumin gene is not expressed.

VISUAL SKILLS *Describe the enhancer for the albumin gene in each type of cell. How would the nucleotide sequence of this enhancer in the liver cell compare with that in the lens cell?*

*For suggested answer, see*Appendix A*.*

Coordinately Controlled Genes in Eukaryotes

How does the eukaryotic cell deal with a group of genes of related function that need to be turned on or off at the same time? Earlier in this chapter, you learned that in bacteria, such *coordinately controlled* genes are often clustered into an operon, which is regulated by a single promoter and transcribed into a single mRNA molecule. Thus, the genes are expressed together, and the encoded proteins are produced at the same time. With a few exceptions, operons that work in this way have *not* been found in eukaryotic cells.

Eukaryotic genes that are co-expressed, such as genes coding for the enzymes of a metabolic pathway, are typically scattered over different chromosomes. Here, coordinate gene expression depends on every gene of a dispersed group having a specific combination of control elements. Transcription activators in the nucleus that recognize the control elements bind to them, promoting simultaneous transcription of the genes, no matter where they are in the genome.

Coordinate control of dispersed genes in a eukaryotic cell often occurs in response to chemical signals from outside the cell. A steroid hormone, for example, enters a cell and binds to a specific intracellular receptor protein, forming a hormone-receptor complex that serves as a transcription activator (see Figure 11.9). Every gene whose transcription is stimulated by a given steroid hormone, regardless of its chromosomal location, has a control element recognized by that hormone-receptor complex. This is how estrogen activates a group of genes that stimulate cell division in uterine cells, preparing the uterus for pregnancy.

Many signaling molecules, such as nonsteroid hormones and growth factors, bind to receptors on a cell's surface and never actually enter the cell. Such molecules can control gene expression indirectly by triggering signal transduction pathways that activate particular transcription factors (see Figure 11.15). Coordinate regulation in such pathways is the same as for steroid hormones: Genes with the same sets of control elements are activated by the same chemical signals. Because this system for coordinating gene regulation is so widespread, biologists think that it probably arose early in evolutionary history.

Nuclear Architecture and Gene Expression

Each chromosome in the interphase nucleus of animal cells occupies a distinct territory (see Figure 16.24a). Recently, *chromosome conformation capture* techniques (3C and others) have been developed that allow researchers to cross-link and identify regions of chromosomes associating with each other during interphase. These studies reveal two organizational details: First, the territory of each chromosome is divided into regions of chromatin loops (*topologically associated domains,* or *TADs*), within which chromatin sites associate mainly with each other. Second, loops of chromatin, each likely a TAD, extend from individual chromosomal territories into specific sites in the nucleus **(Figure 18.13)**. Different loops from the same chromosome and loops from other chromosomes may congregate in such sites, some of which are rich in RNA polymerases and other transcription-associated proteins. Like a recreation center that draws members from many different neighborhoods, these so-called *transcription factories* are thought to be areas specialized for a common function.

Figure 18.13

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Chromosomal interactions in the interphase nucleus.
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Image of chromosomes in the interphase nucleus (fluorescence micrograph, scale: 5 micrometers) depicting splotches of color in distinct regions. Diagram zooms in on region of 3 different colors.

Image of chromosomes in the interphase nucleus (fluorescence micrograph, scale: 5 micrometers) depicting splotches of color in distinct regions. Diagram zooms in on region of 3 different colors. Each color (distinct

region) represents a separate chromosome territory, and chromatin loops extend from each territory towards an overlapping area in the middle of all three territories. The blue chromosome has two chromatin loops; the red and green chromosome territories each have one loop shown.

Although each chromosome has its own territory (see Figure 16.24a), loops of chromatin may extend into other sites in the nucleus. (Each loop may be a topologically associated domain, or TAD.) Some of these sites are transcription factories that are occupied by multiple chromatin loops from the same chromosome (blue loops) or other chromosomes (red and green loops).

The old view that the nuclear contents are like a bowl of amorphous chromosomal spaghetti has given way to a new model of a nucleus with a defined architecture and regulated movements of chromatin. Several lines of evidence suggest that unexpressed genes are located in the outer edges of the nucleus, while those that are being expressed are found in its interior region. Relocation of particular genes from their chromosomal territories to transcription factories in the interior may be part of the process of readying genes for transcription. In 2017, a consortium of researchers funded by the National Institutes of Health began investigating genome organization over time and its relationship to genome function.

Mechanisms of Post-transcriptional Regulation

Transcription alone does not constitute gene expression. The expression of a protein-coding gene is ultimately measured by the amount of functional protein a cell makes, and much happens between the synthesis of the RNA transcript and the activity of the protein in the cell. Many regulatory mechanisms operate at the various stages after transcription (see Figure 18.6). These mechanisms allow a cell to rapidly fine-tune gene expression in response to environmental changes without altering its transcription patterns. Here we explore how cells regulate gene expression after transcription.

RNA Processing

RNA processing in the nucleus and the export of mature RNA to the cytoplasm provide opportunities for regulating gene expression not available in prokaryotes. One example of regulation at the RNA-processing level is **alternative RNA splicing**, in which different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as exons and which as introns. Regulatory proteins specific to a cell type control intron/exon choices by binding to regulatory sequences within the primary transcript.

A simple example of alternative RNA splicing is shown in **Figure 18.14** for the troponin T gene, which encodes multiple closely related proteins with slightly different effects on muscle contraction. Other genes code for many more products. For instance, researchers have found a *Drosophila* gene with enough

alternatively spliced exons to generate about 19,000 membrane proteins that have different extracellular domains. At least 17,500 (94%) of the alternative mRNAs are actually synthesized. Each developing nerve cell in the fly appears to synthesize a different form of the protein, which acts as a unique identifier on the cell surface and helps prevent excessive overlap of nerve cells during development of the nervous system.

Figure 18.14

Alternative RNA splicing of the troponin T gene.

Diagram flows from D N A with Troponin T gene with exon regions labeled to the primary R N A transcript to two different m R N A molecules following R N A splicing.

The primary transcript of this gene can be spliced in more than one way, generating different mRNA molecules. Notice that one mRNA molecule has ended up with exon 3 (green) and the other with exon 4 (purple). These two mRNAs are translated into different but related muscle proteins.

Overview: Stages in gene expression subject to regulation detailed in a flow chart: Within nucleus: Chromatin modification, transcription, R N A processing; in the cytoplasm: m R N A degradation, translation and protein processing and degradation. R N A processing is highlighted for this figure. D N A with Troponin T gene. Exon regions of the gene labeled 1, 2, 3, 5 and 5. Intron regions separate the exons but are not labeled. Following transcription, the primary R N A transcript is made. Diagram shows R N A with regions corresponding to exons 1, 2, 3, 4 and 5. Following R N A splicing, one m R N A molecule carries exons 1, 2, 3 and 5, and the other m R N A carries 1, 2, 4 and 5.

Alternative RNA splicing can significantly expand the repertoire of a eukaryotic genome. In fact, alternative splicing was proposed as one explanation for the surprisingly low number of human genes counted when the human genome was sequenced. The number of human genes was found to be similar to that of a soil worm (nematode), a mustard plant, or a sea anemone. This discovery prompted questions about what, if not the number of genes, accounts for the more complex form and structure of humans. More than 90% of human protein-coding genes likely undergo alternative splicing. Thus, the extent of alternative splicing greatly multiplies the number of possible human proteins, which may be better correlated with complexity of form than the number of genes.

Initiation of Translation and mRNA Degradation

Translation is another stage where gene expression is regulated, most commonly at the initiation stage (see Figure 17.19). For some mRNAs, the initiation of translation can be blocked by regulatory proteins that bind to specific sequences or structures within the untranslated region (UTR) at the 5′ or 3′ end, preventing the attachment of ribosomes. (Recall from Concept 17.3 that both the 5′ cap and the poly-A tail of an mRNA molecule are important for ribosome binding.)

Alternatively, translation of *all* the mRNAs in a cell may be regulated simultaneously. In a eukaryotic cell, such “global” control usually involves the activation or inactivation of one or more protein factors required to initiate translation. This mechanism plays a role in starting translation of mRNAs that are stored in eggs. Just after fertilization, translation is triggered by the sudden activation of translation initiation factors. The response is a burst of synthesis of the proteins encoded by the stored mRNAs. Some plants and algae store mRNAs during periods of darkness; light then triggers the reactivation of the translational apparatus.

The life span of mRNA molecules in the cytoplasm is important in determining the pattern of protein synthesis in a cell. Bacterial mRNA molecules typically are degraded by enzymes within a few minutes. This short life span of mRNAs is one reason bacteria can change their patterns of protein synthesis so quickly in response to environmental changes. In contrast, some mRNAs in multicellular eukaryotes typically survive for hours, days, or even weeks. For instance, the mRNAs for the hemoglobin polypeptides (α-globin and β-globin) in developing red blood cells are unusually stable, and these long-lived mRNAs are translated repeatedly in red blood cells.

Nucleotide sequences that affect how long an mRNA remains intact are often found in the untranslated region at the 3′ end of the molecule (see Figure 18.9). In one experiment, researchers transferred such a sequence from the short-lived mRNA for a growth factor to the 3′ end of a normally stable globin mRNA. The globin mRNA was quickly degraded.

Other mechanisms that degrade or block expression of mRNA molecules have come to light. They involve a group of RNA molecules that regulate gene expression at several levels, as you’ll see shortly.

Protein Processing and Degradation

The final opportunities for controlling gene expression occur after translation. Often, eukaryotic polypeptides must be processed to yield functional protein molecules. For instance, cleavage of the initial insulin polypeptide (pro-insulin) forms the active hormone. In addition, many proteins undergo chemical modifications that make them functional. Regulatory proteins are commonly activated or inactivated by the reversible addition of phosphate groups (see Figure 11.10), and proteins destined for the surface of animal cells acquire sugars (see Figure 6.12). Cell-surface proteins and many others must also be transported to target destinations in the cell in order to function (see Figure 17.22). Regulation might occur at any of the steps involved in modifying or transporting a protein.

Finally, the length of time each protein functions in the cell is strictly regulated by selective degradation. Many proteins, such as the cyclins involved in regulating the cell cycle, must be relatively short-lived if the cell is to function appropriately (see Figure 12.16). To mark a protein for destruction, the cell commonly attaches molecules of a small protein called ubiquitin to the protein. Giant protein complexes called proteasomes then recognize the ubiquitin tagged proteins and degrade them.

Animation

Post-Transcriptional Control Mechanisms

image

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